

Shared signals and the potential for phylogenetic espionage between plants and animals.

Until recently, the study and understanding of plant and animal signalling and response mechanisms have developed independently. Recent biochemical and molecular work is producing a growing list of elements involved in responses to biotic and abiotic stimuli that are very similar across kingdoms. Some of the more interesting examples of these include prostaglandin/octadecanoid-mediated responses to wounding, steroid-based signalling systems, and pathogen-recognition mechanisms. Some of these similarities probably represent evolutionary convergence; others may be ancestral to plants and animals. Ecological and evolutionary implications of such overlaps include the existence of pathogens that can cause disease in plants and animals, the ability of herbivores to manipulate plant responses, usurpation of microbial mechanisms and genes by herbivorous animals and plants, evolution of plant defenses exploiting shared signals in animals, and the medicinal use of plants by humans. Comparative study of the signalling and response mechanisms used by plants, animals, and microbes provides novel and useful insights to the ecology and evolution of interactions across kingdoms.